

EShop HW04 - Multi-browser Automation Report

Run by

23127216

1. Scope & Oracle

Item	Value
SUT	EShop (apps/ monorepo) — storefront :5173 , admin :5174 , API :3000
Features under test	FR-02 Login & Account Lockout · FR-09 Discount Coupons · FR-16 CSV Import
Test design source	HW02 domain-testing.md + bva.md per feature, re-verified against server.js for HW04
Technique	Domain Testing (Equivalence Partitioning) + Boundary Value Analysis, automated with Playwright
Test type	Data-driven, multi-browser (Chromium / Firefox / Edge — Edge substitutes for WebKit, see bugs/BUG_REPORT.md), one test() per case in test-data/<FR>/cases.json
Oracle	The spec (docs/eshop-sut/srs.md , api_specification.md) — NOT the SUT's current implementation. A discrepancy between spec and implementation is a bug, not a reason to change the expected value.

1.1 Why a cross-browser matrix, not a single-column table

Each test case runs on 3 browsers. A case can pass on Chromium and fail on Edge for reasons unrelated to the feature under test (HTML5 validation differences, number formatting, file input handling) — this is itself a defect worth reporting, and HW02's manual, single-pass testing could not surface it. §3–5 present results as **TC × browser** matrices instead of one PASS/FAIL column, so any row with disagreeing cells is visible without diffing 3 separate HTML reports by hand.

1.2 Report layers

Layer	Purpose	Location
1. Playwright HTML report (native)	Deep-dive: trace, screenshot, timeline for one case	html-reports/<FR>/<browser>/index.html
2. JSON reporter	Machine-readable input for layer 3	test-results/<FR>/<browser>/results.json
3. This PDF	Single-file, human-first view: summary + per-feature matrix + bug index	pdf/HW04_Test_Report.pdf

2. Summary

63 test cases across 3 features, run on Chromium / Firefox / Edge (up to 189 runs). **30 case(s)** deviate from spec on at least one browser. **0 case(s)** diverge between browsers — see §3–5 for the per-feature matrix.

Feature	TC	Failed (≥ 1 browser)	Browser-diff
FR-02	18	8	0
FR-09	21	10	0
FR-16	24	12	0
Total	63	30	0

3. Results — FR-02

18 test cases × 3 browsers = 54 runs planned. **0 case(s) show cross-browser divergence** (bold cell +  browser-diff) — these are automation-only findings HW02's manual pass could not produce.

TC	Layer	Expected (spec)	Chromium	Firefox	Edge	Known Defect
TC-01	hybrid	UI: {"noErrorVisible":true,"navigatesAwayFromLogin":true}; API: {"status":200,"hasToken":true}; DB: {"login_attempts":0,"locked_until":null}	PASS	PASS	PASS	
TC-02	ui	UI: {"staysOnLoginPage":true,"html5ValidationOnUsername":true}; API: {"requestSent":false}	PASS	PASS	PASS	
TC-03	ui	UI: {"staysOnLoginPage":true,"html5ValidationOnUsername":true}; API: {"requestSent":false}	FAIL	FAIL	FAIL	BUG-02-001
TC-03b	ui	UI: {"staysOnLoginPage":true,"html5ValidationOnUsername":true}; API: {"requestSent":false}	FAIL	FAIL	FAIL	BUG-02-001
TC-03c	ui	UI: {"staysOnLoginPage":true,"html5ValidationOnUsername":true}; API: {"requestSent":false}	FAIL	FAIL	FAIL	BUG-02-001
TC-03d	ui	UI: {"staysOnLoginPage":true,"html5ValidationOnUsername":true}; API: {"requestSent":false}	FAIL	FAIL	FAIL	BUG-02-001
TC-04	hybrid	UI: {"genericErrorVisible":true,"errorRevealsReason":false}; API: {"status":401,"hasToken":false}	PASS	PASS	PASS	
TC-05	hybrid	UI: {"staysOnLoginPage":true}; API: {"requestSent":false}; DB: {"login_attempts":0}	PASS	PASS	PASS	
TC-06	hybrid	UI: {"genericErrorVisible":true,"errorRevealsReason":false}; API: {"status":401,"hasToken":false}; DB: {"login_attempts":1}	FAIL	FAIL	FAIL	BUG-02-003
TC-07	hybrid	UI: {"genericErrorVisible":true,"errorRevealsReason":false}; API: {"hasToken":false,"status":401}	FAIL	FAIL	FAIL	PENDING-NEW-lockout-message-enumeration
TC-BVA-01	hybrid	API: {"hasToken":false}; DB: {"login_attempts":3,"locked_untilNotNull":true}	FAIL	FAIL	FAIL	BUG-02-003
TC-BVA-02	hybrid	API: {"status":200,"hasToken":true}; DB: {"login_attempts":0,"locked_until":null}	PASS	PASS	PASS	
TC-BVA-03	hybrid	UI: {"genericErrorVisible":true}; API: {"hasToken":false}	PASS	PASS	PASS	
TC-BVA-04	hybrid	UI: {"noErrorVisible":true}; API: {"status":200,"hasToken":true}; DB: {"login_attempts":0,"locked_until":null}	PASS	PASS	PASS	
TC-BVA-05	hybrid	UI: {"noErrorVisible":true}; API: {"status":200,"hasToken":true}; DB: {"login_attempts":0,"locked_until":null}	PASS	PASS	PASS	
TC-UI-01	ui	UI: {"passwordInputType":"password"}	FAIL	FAIL	FAIL	PENDING-NEW-password-field-not-masked
TC-UI-02	ui	UI: {"noErrorVisible":true,"navigatesAwayFromLogin":true}; API: {"status":200,"hasToken":true}	PASS	PASS	PASS	
TC-UI-03	ui	UI: {"genericErrorVisible":true,"errorRevealsReason":false}	PASS	PASS	PASS	PENDING-NEW-lockout-message-enumeration

4. Results — FR-09

21 test cases × 3 browsers = 63 runs planned. **0 case(s) show cross-browser divergence** (bold cell +  browser-diff) — these are automation-only findings HW02's manual pass could not produce.

TC	Layer	Expected (spec)	Chromium	Firefox	Edge	Known Defect
TC-01	hybrid	API: {"status":200,"discount_amount":50000,"final_amount":450000}	FAIL	FAIL	FAIL	BUG-09-001
TC-02	hybrid	API: {"status":200,"discount_amount":50000,"final_amount":550000}	PASS	PASS	PASS	
TC-03	hybrid	API: {"status":404,"hasError":true,"hasDiscount":false}	PASS	PASS	PASS	
TC-04	hybrid	API: {"status":404,"hasError":true,"hasDiscount":false}	PASS	PASS	PASS	
TC-05	hybrid	API: {"status":404,"hasError":true,"hasDiscount":false}	FAIL	FAIL	FAIL	BUG-09-006
TC-06	hybrid	API: {"status":400,"hasError":true,"hasDiscount":false}	PASS	PASS	PASS	
TC-07	hybrid	API: {"status":400,"hasError":true,"hasDiscount":false}	PASS	PASS	PASS	
TC-08	api	API: {"status":401,"hasError":true,"hasDiscount":false}	FAIL	FAIL	FAIL	BUG-09-002
TC-09	api	API: {"status":403,"hasError":true,"hasDiscount":false}	FAIL	FAIL	FAIL	BUG-09-003
TC-10	hybrid	API: {"status":400,"hasError":true,"hasDiscount":false}	PASS	PASS	PASS	
TC-11	hybrid	API: {"finalAmountNotNegative":true}	FAIL	FAIL	FAIL	BUG-09-004
TC-12	hybrid	API: {"status":200,"discount_amount":0,"final_amount":0}	FAIL	FAIL	FAIL	BUG-09-005
TC-13	hybrid	API: {"status":200,"hasDiscount":true}; DB: {"orderTotalMatchesRealSubtotal":true}	FAIL	FAIL	FAIL	BUG-09-007
TC-BVA-01	hybrid	API: {"status":400,"hasError":true,"hasDiscount":false}	PASS	PASS	PASS	
TC-BVA-02	hybrid	API: {"status":200,"discount_amount":30000,"final_amount":270000}	FAIL	FAIL	FAIL	BUG-09-005
TC-BVA-03	hybrid	API: {"status":200,"discount_amount":30000,"final_amount":270001}	FAIL	FAIL	FAIL	BUG-09-001
TC-BVA-04	hybrid	API: {"status":200,"discount_amount":100000,"final_amount":300000}	PASS	PASS	PASS	
TC-BVA-05	hybrid	API: {"status":400,"hasError":true,"hasDiscount":false}	PASS	PASS	PASS	
TC-BVA-06	hybrid	API: {"status":400,"hasError":true,"hasDiscount":false}	PASS	PASS	PASS	
TC-BVA-07	hybrid	API: {"status":400,"hasError":true,"hasDiscount":false}	PASS	PASS	PASS	
TC-BVA-08	hybrid	API: {"status":200,"discount_amount":50000,"final_amount":450000}	FAIL	FAIL	FAIL	BUG-09-001

5. Results — FR-16

24 test cases × 3 browsers = 72 runs planned. **0 case(s) show cross-browser divergence** (bold cell +  browser-diff) — these are automation-only findings HW02's manual pass could not produce.

TC	Layer	Expected (spec)	Chromium	Firefox	Edge	Known Defect
TC-01	hybrid	UI: {"previewCount":1,"resultTextIncludes":"1/1","errorsCount":0}; API: {"status":200,"inserted":1,"errorsCount":0}; DB: {"productExists":"Basic Men's T-Shirt","productCountDelta":1,"persistedRowsMatchInput":true}	PASS	PASS	PASS	
TC-02	hybrid	UI: {"previewCount":3,"resultTextIncludes":"3/3","errorsCount":0}; API: {"status":200,"inserted":3,"errorsCount":0}; DB: {"productCountDelta":3,"persistedRowsMatchInput":true}	PASS	PASS	PASS	
TC-03	api	API: {"status":401,"hasError":true}; DB: {"productCountDelta":0,"productsNotCreated":["Test"]}	PASS	PASS	PASS	
TC-04	api	API: {"status":403,"hasError":true}; DB: {"productCountDelta":0,"productsNotCreated":["Test"]}	FAIL	FAIL	FAIL	BUG-16-001
TC-05	api	API: {"status":400,"hasError":true}; DB: {"productCountDelta":0}	PASS	PASS	PASS	
TC-06	hybrid	UI: {"previewCount":1,"resultTextIncludes":"0/1","errorsCount":1}; API: {"inserted":0,"errorsCount":1,"errorMentions":"name"}; DB: {"productCountDelta":0}	PASS	PASS	PASS	
TC-07	hybrid	UI: {"previewCount":1,"resultTextIncludes":"0/1","errorsCount":1}; API: {"inserted":0,"errorsCount":1,"errorMentions":"name"}; DB: {"productCountDelta":0}	PASS	PASS	PASS	
TC-08	hybrid	UI: {"previewCount":1,"resultTextIncludes":"0/1","errorsCount":1}; API: {"inserted":0,"errorsCount":1,"errorMentions":"price"}; DB: {"productCountDelta":0,"productsNotCreated":["Test zero price"]}	FAIL	FAIL	FAIL	BUG-16-002
TC-09	hybrid	UI: {"previewCount":1,"resultTextIncludes":"0/1","errorsCount":1}; API: {"inserted":0,"errorsCount":1,"errorMentions":"price"}; DB: {"productCountDelta":0,"productsNotCreated":["Test negative price"]}	FAIL	FAIL	FAIL	BUG-16-002
TC-10	hybrid	UI: {"previewCount":1,"resultTextIncludes":"0/1","errorsCount":1}; API: {"inserted":0,"errorsCount":1,"errorMentions":"price"}; DB: {"productCountDelta":0,"productsNotCreated":["Test bad price"]}	FAIL	FAIL	FAIL	BUG-16-002
TC-11	hybrid	UI: {"previewCount":1,"resultTextIncludes":"0/1","errorsCount":1}; API: {"inserted":0,"errorsCount":1,"errorMentions":"price"}; DB: {"productCountDelta":0,"productsNotCreated":["Test no price"]}	FAIL	FAIL	FAIL	BUG-16-002
TC-12	hybrid	UI: {"previewCount":3,"resultTextIncludes":"0/3","errorsCount":1,"errorRows": [3],"errorRowsMention":"price"}; API: {"inserted":0,"errorsCount":1,"errorRows": [3],"errorRowsMention":"price"}; DB: {"productCountDelta":0,"productsNotCreated":["Valid Product 1","Valid Product 2"]}	FAIL	FAIL	FAIL	BUG-16-002
TC-13	api	API: {"status":400,"hasError":true}; DB: {"productCountDelta":0}	PASS	PASS	PASS	
TC-14	hybrid	UI: {"previewCount":1,"resultTextIncludes":"0/1","errorsCount":1}; API: {"inserted":0,"errorsCount":1}; DB: {"productCountDelta":0}	FAIL	FAIL	FAIL	BUG-16-004
TC-15	hybrid	UI: {"previewCount":1,"resultTextIncludes":"0/1","errorsCount":1}; API: {"inserted":0,"errorsCount":1}; DB: {"productCountDelta":0,"productsNotCreated":["Test invalid category"]}	FAIL	FAIL	FAIL	BUG-16-003
TC-16	hybrid	UI: {"previewCount":1,"resultTextIncludes":"1/1","errorsCount":0}; API: {"status":200,"inserted":1}; DB: {"productExists":"Test price as string","productCountDelta":1,"persistedRowsMatchInput":true}	PASS	PASS	PASS	
TC-17	api	API: {"status":200,"hasFields":["message","inserted","errors"],"fieldTypes":{"message":"string","inserted":"number","errors":"array"},"inserted":1}; DB: {"productCountDelta":1,"persistedRowsMatchInput":true}	PASS	PASS	PASS	
TC-18	hybrid	UI: {"previewCount":3,"resultTextIncludes":"0/3","errorsCount":3,"errorRows": [2,3,4],"errorRowsMention":"price"}; API: {"inserted":0,"errorsCount":3,"errorRows": [2,3,4],"errorRowsMention":"price"}; DB: {"productCountDelta":0,"productsNotCreated":["Invalid Row 1","Invalid Row 2","Invalid Row 3"]}	FAIL	FAIL	FAIL	BUG-16-002
TC-19	hybrid	UI: {"acceptCsvOnly":true,"fileRejectedBeforeUpload":true,"validResultTextIncludes":"1/1"};	FAIL	FAIL	FAIL	BUG-16-005

TC	Layer	Expected (spec)	Chromium	Firefox	Edge	Known Defect
		API: {"status":200,"requestContentType":"application/json","requestProductsCount":1}; DB: {"productCountDelta":1,"persistedRowsMatchInput":true}				
TC-BVA-01	hybrid	UI: {"previewCount":1,"resultTextIncludes":"0/1","errorsCount":1}; API: {"inserted":0,"errorsCount":1}; DB: {"productCountDelta":0,"productsNotCreated":["BVA price zero"]}	FAIL	FAIL	FAIL	BUG-16-002
TC-BVA-02	hybrid	UI: {"previewCount":1,"resultTextIncludes":"1/1","errorsCount":0}; API: {"status":200,"inserted":1}; DB: {"productExists":"BVA price one","productCountDelta":1,"persistedRowsMatchInput":true}	PASS	PASS	PASS	
TC-BVA-03	hybrid	UI: {"previewCount":1,"resultTextIncludes":"1/1","errorsCount":0}; API: {"status":200,"inserted":1}; DB: {"productCountDelta":1,"productNameLength":254,"persistedRowsMatchInput":true}	PASS	PASS	PASS	
TC-BVA-04	hybrid	UI: {"previewCount":1,"resultTextIncludes":"1/1","errorsCount":0}; API: {"status":200,"inserted":1}; DB: {"productCountDelta":1,"productNameLength":255,"persistedRowsMatchInput":true}	PASS	PASS	PASS	
TC-BVA-05	hybrid	UI: {"previewCount":1,"resultTextIncludes":"0/1","errorsCount":1}; API: {"inserted":0,"errorsCount":1}; DB: {"productCountDelta":0}	FAIL	FAIL	FAIL	BUG-16-004

6. Bug Cross-Reference

Auto-generated from `knownDefect` fields in `test-data/<FR>/cases.json`. Full narrative, severity, and GitHub Issue links live in `../../../../bugs/BUG_REPORT.md` — this section is a quick index, not a replacement.

FR-02

TC	Known Defect	Title
TC-03	BUG-02-001	Invalid email format (no @ sign) blocked by HTML5 type=email validation
TC-03b	BUG-02-001	Invalid email format (missing domain, e.g. "test@") blocked by HTML5 validation
TC-03c	BUG-02-001	Invalid email format (missing local-part, e.g. "@domain.com") blocked by HTML5 validation
TC-03d	BUG-02-001	Invalid email format (embedded space, e.g. "test @domain.com") blocked by HTML5 validation
TC-06	BUG-02-003	Wrong password increments counter by exactly 1
TC-07	PENDING-NEW-lockout-message-enumeration	Locked account rejects even correct credentials
TC-BVA-01	BUG-02-003	3rd failure (2->3) triggers lock immediately (ON point)
TC-UI-01	PENDING-NEW-password-field-not-masked	Password field must mask input (type=password)
TC-UI-03	PENDING-NEW-lockout-message-enumeration	After lock, UI reflects lockout state, not only the API

FR-09

TC	Known Defect	Title
TC-01	BUG-09-001	Apply valid percent coupon (happy path)
TC-05	BUG-09-006	Coupon entered with wrong case is rejected (case-sensitivity gap)
TC-08	BUG-09-002	No Authorization header must be rejected
TC-09	BUG-09-003	Invalid JWT token must be rejected
TC-11	BUG-09-004	Gap: fixed discount_value greater than total_amount must not yield a negative final_amount
TC-12	BUG-09-005	Gap: total_amount=0 with min_order_amount=0 is a valid degenerate case (>=)
TC-13	BUG-09-007	Gap: client-manipulated total_amount must not let the persisted order diverge from the real cart subtotal
TC-BVA-02	BUG-09-005	total_amount exactly at min_order_amount must be accepted (ON point, >=)
TC-BVA-03	BUG-09-001	total_amount one unit above min_order_amount is accepted (UB+1)
TC-BVA-08	BUG-09-001	expired_at = tomorrow (UB+1) is still valid

FR-16

TC	Known Defect	Title
TC-04	BUG-16-001	Valid JWT but not an admin must be rejected
TC-08	BUG-16-002	price = 0 (not positive) must reject the row
TC-09	BUG-16-002	Negative price must reject the row
TC-10	BUG-16-002	Non-numeric price string must reject the row
TC-11	BUG-16-002	price field absent must reject the row
TC-12	BUG-16-002	Atomic rollback: 1 invalid row among 3 must roll back the entire batch
TC-14	BUG-16-004	Gap Probe: name longer than 255 characters (FR-15 cross-ref limit)
TC-15	BUG-16-003	Gap Probe: category_id does not exist in DB (dangling FK)
TC-18	BUG-16-002	Gap Probe: all-rows-invalid batch must list a distinct reason per row
TC-19	BUG-16-005	Gap Probe: Admin UI must enforce the .csv file extension
TC-BVA-01	BUG-16-002	price = 0 at the ON point must be rejected
TC-BVA-05	BUG-16-004	name = 256 chars (UB+1) must be rejected if the 255-char limit is enforced

7. Notes & How to Reproduce

7.1 Reading the matrix (§3–5)

- **PASS / FAIL** — verdict of that TC on that browser, from the Playwright JSON reporter (`spec.ok`).
- **Bold cell** — this row's cells disagree across browsers (`▲ browser-diff` in the last column). Investigate before assuming it is "just" the known defect in the `Known Defect` column — a browser-diff can stack on top of a spec bug.
- `–` — that browser's `results.json` was not found (run not executed yet, or run failed to produce a report).
- **Known Defect** — copied from `knownDefect` in `test-data/<FR>/cases.json` . `null` /empty means the case is expected to PASS; anything else names the bug this case's FAIL is expected evidence for.

7.2 Verdict notes

- Expected values in `test-data/<FR>/cases.json` are written against **spec**, not against the SUT's observed behaviour (see §1). A FAIL against a `knownDefect` case is the intended, correct outcome — it is evidence, not suite breakage.
- `retries: 0` in `playwright.config.ts` — no automatic retry, so a FAIL is never masked by a lucky rerun.

7.3 How to regenerate this report

```
cd homeworks/HW04/artifacts
npm install
npx playwright install chromium firefox
flatpak install --user flathub-user com.microsoft.Edge # Edge substitutes for WebKit (see bugs/BUG_REPORT.md)

# terminal khác:
bash ../../../../apps/run-servers.sh # :3000 / :5173 / :5174

# 1 run = 1 feature x 1 browser, JSON + HTML both written:
STUDENT_ID=23127216 \
HTML_OUT=html-reports/FR-02/chromium \
JSON_OUT=test-results/FR-02/chromium/results.json \
npx playwright test tests/FR-02 --project=chromium
# repeat for FR-09, FR-16 x chromium, firefox, edge (9 runs total)

npm run report:summary # reads all test-results/*/*/results.json -> reports/markdown/02,03,04,05,06_*.md
npm run report:pdf # concatenates reports/markdown/*.md -> pdf/HW04_Test_Report.pdf
```

Artifacts produced

Path	Contents
<code>test-data/<FR>/cases.json</code>	Data-driven test cases (source of truth for expected + knownDefect)
<code>tests/<FR>/*.spec.ts</code>	Data-driven Playwright specs, one loop per feature
<code>html-reports/<FR>/<browser>/</code>	Interactive Playwright HTML report (9 folders)
<code>test-results/<FR>/<browser>/results.json</code>	Machine-readable results (9 files)
<code>reports/markdown/</code>	This report's Markdown source (auto + hand-written sections)
<code>pdf/HW04_Test_Report.pdf</code>	Rendered, self-contained PDF — the single-file view